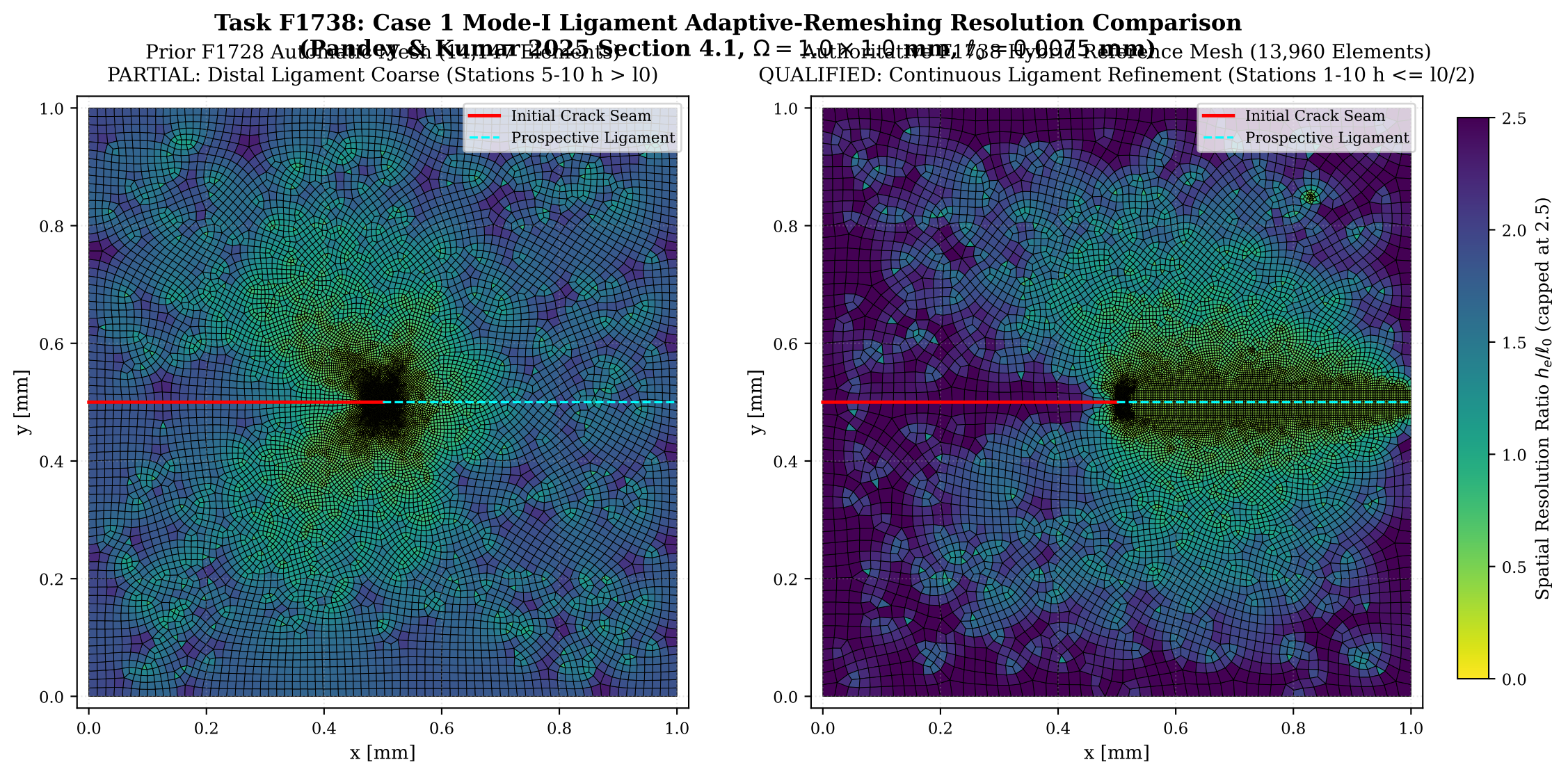

Task F1738: Case 1 Mode-I Ligament Adaptive-Remeshing Resolution Comparison
(Pandey & Kumar 2025 Section 4.1, Ω = 1.0 × 1.0 mm, ℓ = 0.0075 mm)
Prior F1728 Automatic Mesh (14,144 Elements)
PARTIAL: Distal Ligament Coarse (Stations 5-10 h > l0)
Adaptive F1738 Hybrid Reference Mesh (13,960 Elements)
QUALIFIED: Continuous Ligament Refinement (Stations 1-10 h <= l0/2)
Initial Crack Seam
Prospective Ligament
y [mm]
x [mm]
Spatial Resolution Ratio $h_e/\ell_0$ (capped at 2.5)
0.0
0.2
0.4
0.6
0.8
1.0
0.5
1.5
2.0
2.5